Text Summarization System Report

TF-IDF Extractive and Transformer-Based Abstractive Summarization

1. Problem Description

Text summarization is one of the most important applications in Natural Language Processing (NLP). The main goal is to reduce a long text into a shorter version while preserving the most important information and main ideas. This project implements a complete text summarization system using two different approaches:

1. **Extractive Summarization** — Baseline Method
2. **Abstractive Summarization** — Advanced Method

The extractive model selects the most important sentences directly from the original text using TF-IDF scoring and sentence embeddings. The abstractive model uses a Transformer-based architecture, specifically the BART model, which generates new fluent summaries and can paraphrase the original text.

The system was designed to:

- Keep important information from the original article
- Reduce the length of the text significantly
- Preserve the meaning of the original article
- Compare traditional and advanced summarization techniques

2. Dataset Used

The project uses articles and their corresponding summaries for training and evaluation. The system is suitable for various types of text including:

- News Articles
- Blog Posts
- Product Reviews
- General Long Documents

The abstractive model uses **facebook/bart-large-cnn**, which was originally trained on the **CNN/DailyMail summarization dataset**. The dataset details are:

- **Source:** News articles from CNN and DailyMail
- **Size:** 5000 samples (configurable)
- **Split:** 80% train, 10% validation, 10% test
- **Article Length:** ~700 words on average
- **Highlights Length:** ~50 words on average

Evaluation was performed on multiple text samples to compare the generated summaries from both the extractive and abstractive methods.

3. Preprocessing Steps

Before training and summarization, the text goes through several preprocessing steps:

1. Convert text to lowercase

Example: "The Food Was Amazing" → "the food was amazing"

2. Remove punctuation

Symbols such as commas, periods, and special characters are removed to focus on word content.

3. Remove stopwords

Common words like "the", "is", and "and" are removed because they do not carry important meaning.

4. Tokenization

Text is split into individual words and sentences for processing.

5. Sentence Segmentation

Articles are divided into individual sentences using the NLTK sentence tokenizer. This is critical for extractive summarization where we rank and select sentences.

These preprocessing steps improve feature extraction and summarization quality.

4. Feature Extraction

Two feature extraction techniques are used to evaluate sentence importance:

A) TF-IDF — Term Frequency – Inverse Document Frequency

TF-IDF measures how important a word is in a sentence compared to the whole document. Words with higher TF-IDF scores are considered more informative and relevant to the topic.

Advantages:

- Simple and fast computation
- Effective for extractive summarization
- Easy to interpret and explain

B) Sentence Embeddings

Sentence embeddings convert sentences into dense numerical vectors that represent semantic meaning. The model used is **all-MiniLM-L6-v2**, which produces 384-dimensional vectors.

Advantages:

• Captures semantic similarity between sentences
• Understands sentence meaning beyond keyword matching
• Improves sentence ranking quality significantly

5. Baseline Method: Extractive Summarization

The baseline approach uses extractive summarization, which selects the most important sentences directly from the original article.

How it works:

1. Split the article into individual sentences.
2. Calculate TF-IDF scores for each sentence.
3. Generate sentence embeddings using all-MiniLM-L6-v2.
4. Compute cosine similarity between each sentence and the document representation.
5. Combine TF-IDF score and embedding score using weighted average.
6. Select the top-k most important sentences.
7. Restore the original sentence order to produce the final summary.

Formula Used:

$$\textbf{Final Score} = 0.5 \times \textbf{TF-IDF Score} + 0.5 \times \textbf{Embedding Similarity}$$

Advantages:

• Fast execution (~25 seconds for 200 articles)
• Preserves original wording exactly
• No hallucination risk
• Easy to debug and interpret

Disadvantages:

• Cannot paraphrase or generate new text
• Sometimes produces disconnected summaries

6. Advanced Method: Transformer-Based Summarization

The advanced approach uses abstractive summarization with the BART model, which generates entirely new summaries rather than extracting sentences.

Model Used:

facebook/bart-large-cnn (406M parameters)

How it works:

1. The input article is tokenized using the BART tokenizer (max 1024 tokens).

2. The article is encoded into contextual representations by the encoder.
3. The decoder generates a new summary word-by-word.
4. Beam search is used to improve summary quality.

Generation Parameters:

- **Beam Search:** 4
- **Length Penalty:** 1.0 (reduced from 2.0 for more natural summaries)
- **No Repeat N-Gram Size:** 4 (increased from 3)
- **Early Stopping:** Disabled (allows full generation)
- **Max Length:** 130 tokens (~80-100 words)
- **Min Length:** 30 tokens (~15-20 words)

Advantages:

- Generates human-like, fluent summaries
- Can paraphrase and combine ideas from different parts
- Higher ROUGE scores than extractive (~35% better on ROUGE-1)
- Produces more concise summaries (8.6% compression vs 18.9%)

Disadvantages:

- Slower than extractive methods (~505 seconds for 200 articles)
- Requires GPU for reasonable speed
- May occasionally generate incorrect facts (hallucination)

7. Evaluation and Comparison

Models were evaluated using ROUGE metrics, which measure n-gram overlap between generated summaries and human-written reference summaries.

- **ROUGE-1:** Measures unigram (single word) overlap
- **ROUGE-2:** Measures bigram (two-word sequence) overlap
- **ROUGE-L:** Measures longest common subsequence similarity

Additionally, **BERTScore** was used to measure semantic similarity using contextual embeddings, capturing meaning beyond exact word overlap.

Model	ROUGE-1	ROUGE-2	ROUGE-L	Compression Ratio
Extractive	0.2902	0.0991	0.1878	0.1889
Abstractive (BART)	0.3907	0.1692	0.2836	0.0864

Results Analysis:

- The abstractive model achieved better ROUGE scores than the extractive model across all metrics.
- The extractive model was significantly faster (25s vs 505s for 200 samples).
- The abstractive model generated more fluent and natural summaries.
- The extractive model preserved exact wording from the original article.

8. Manual Evaluation

Manual evaluation was performed to verify summary quality beyond automated metrics.

Manual evaluation criteria:

- Whether the summary preserves the main idea
- Whether important information exists in the summary
- Whether the summary is shorter than the original text
- Readability and fluency of the summary

Example:

Original Text:

"The food was amazing and the service was excellent. The restaurant was clean and the staff were friendly. However, the delivery was very slow and the order arrived late."

Extractive Summary:

"The restaurant was clean and the staff were friendly."

Abstractive Summary:

"The food and service were good, but the delivery was slow."

The abstractive summary preserves the important information (food quality, service, delivery issue) while remaining short and readable. The extractive summary selected a single sentence that misses the key point about food quality and the delivery problem.

9. Conclusion

This project implemented a complete text summarization system using both traditional and deep learning techniques. The extractive summarization approach uses TF-IDF and sentence embeddings to select the most important sentences, providing fast and interpretable results. The Transformer-based abstractive summarization using BART generates more fluent, human-like summaries with higher ROUGE scores.

Key findings from this project:

- **Extractive summarization** is suitable when speed and factual accuracy are critical. It preserves the original wording and has no hallucination risk.
- **Abstractive summarization** is suitable for readability and advanced NLP applications. It produces fluent summaries that can paraphrase and synthesize information.

• The project demonstrates how NLP techniques and Transformer models can be combined to build powerful summarization systems capable of reducing long texts while preserving essential meaning.

End of Report